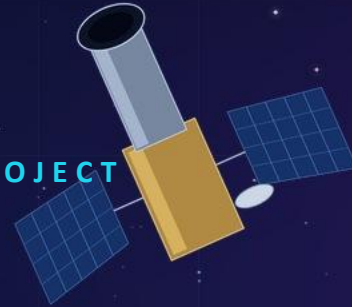


NASA KEPLER EXOPLANET SEARCH RESULTS · HACKATHON PROJECT

# AI-Assisted Kepler Exoplanet Analysis Platform

Exploring Exoplanet Candidates with AI, Machine  
Learning & Data Visualization

TEAM Hack vok



# The Problem & The Dataset



## THE PROBLEM

NASA's Kepler mission gathered a huge amount of information about stars and possible planets beyond our Solar System.

- Thousands of records
- Many technical parameters per record
- Hard to understand what each measurement means
- Manual searching is time-consuming
- Users know what they want, but not which columns to use
- Raw numerical data is difficult to interpret



## THE DATASET

### NASA Kepler Exoplanet Search Results

Selected parameters: planet candidate, transit and signal, host star

#### PLANET / CANDIDATE

Radius

Orbital period

Est. temperature

#### TRANSIT & SIGNAL

Transit duration

Transit depth

Signal-to-noise

#### HOST STAR

Temperature

Radius

Surface gravity

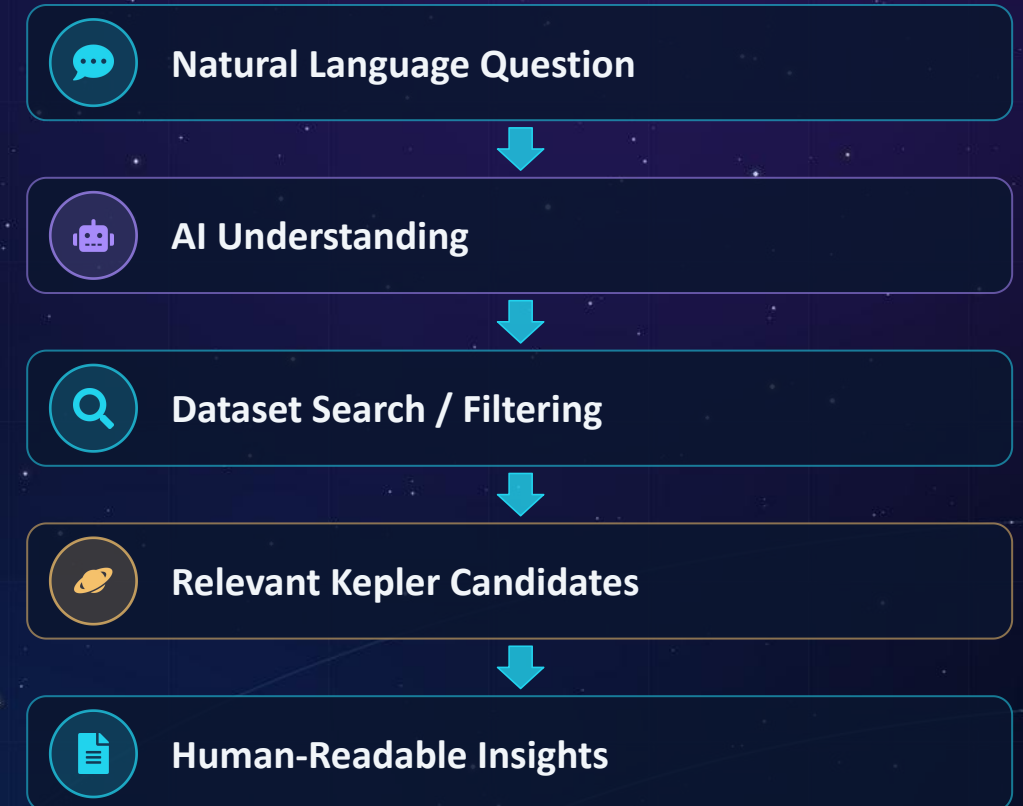
Brightness



Source: NASA Kepler Exoplanet Search Results — Kaggle

# Our Solution

“An Interactive AI-Assisted Kepler Exoplanet Analysis Platform”



Natural-language interaction

Dataset search & filtering

Candidate information

Host-star information

Human-readable explanations

# System Workflow



Conceptual workflow of the proposed platform

# Machine Learning & Candidate Screening



## THE MODELS

Logistic Regression

Random Forest

- Learn patterns from previously classified Kepler objects
- Use multiple parameters together
- Classify candidates from learned patterns

## CANDIDATE SCREENING

 **High Priority**  
Highest follow-up priority

 **Medium Priority**  
Moderate follow-up priority

 **Low Priority**  
Lower follow-up priority

*Assigned using the implemented screening criteria.*

## FEATURE IMPORTANCE

koi\_period

koi\_prad

koi\_depth

koi\_model\_snr

koi\_teq

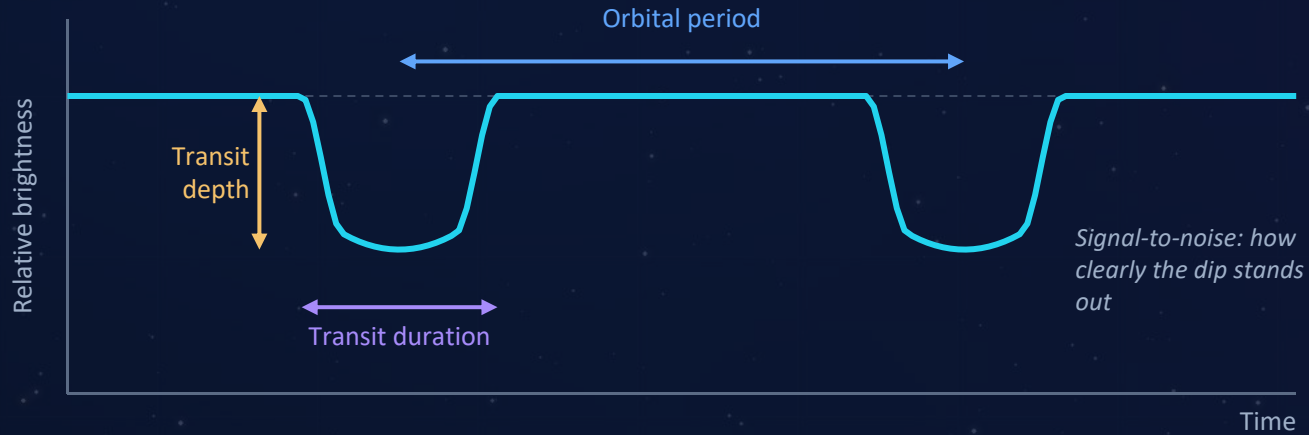
*Conceptual layout — not model output. Actual importances come from the trained model.*



This screening prioritizes candidates for further investigation; it does not scientifically confirm whether a planet is real or false.

# Understanding Transit Events

## TRANSIT LIGHT CURVE · CONCEPTUAL



A planet crossing in front of its star briefly dims the star's light



### Brightness change

Transit depth: how much the star dims



### Duration

How long the transit lasted



### Frequency

How often it occurs (orbital period)



### Signal strength

How strong the detected signal was

## TRANSIT TIMING

$$\text{Expected transit} = \text{reference transit time} + n \times \text{orbital period}$$

When timing and orbital-period information is available, the system can calculate an expected future transit time. **This is a calculation based on available data, not a claim that ML is predicting an entirely new planetary event.**

## VISUALIZATION CONCEPTS

Transit light curve

Distribution plot

Correlation view

Feature importance

Model comparison

Screening results

# From Raw Data to Understandable Insights

## BEFORE · RAW VALUES

```
koi_prad = 1.2  
koi_period = 25.4  
koi_teq = 280
```



## AFTER · HUMAN-READABLE

*"This candidate has a radius of approximately 1.2 Earth radii and completes an orbit around its host star approximately every 25.4 days. Its estimated equilibrium temperature is around 280 K."*



## Astronomers & Exoplanet Researchers

Search

Analyze

Understand

Classify

Prioritize

Kepler candidates — without manually going through the entire dataset.



## Students & Researchers

Explore:

Exoplanet properties

Stellar properties

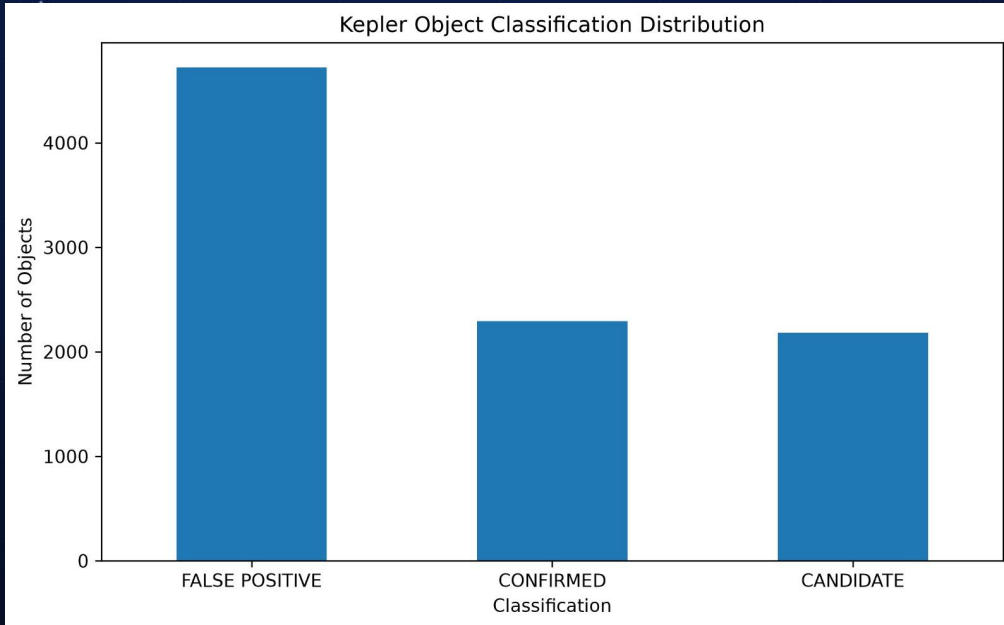
Transit events

Data-analysis techniques

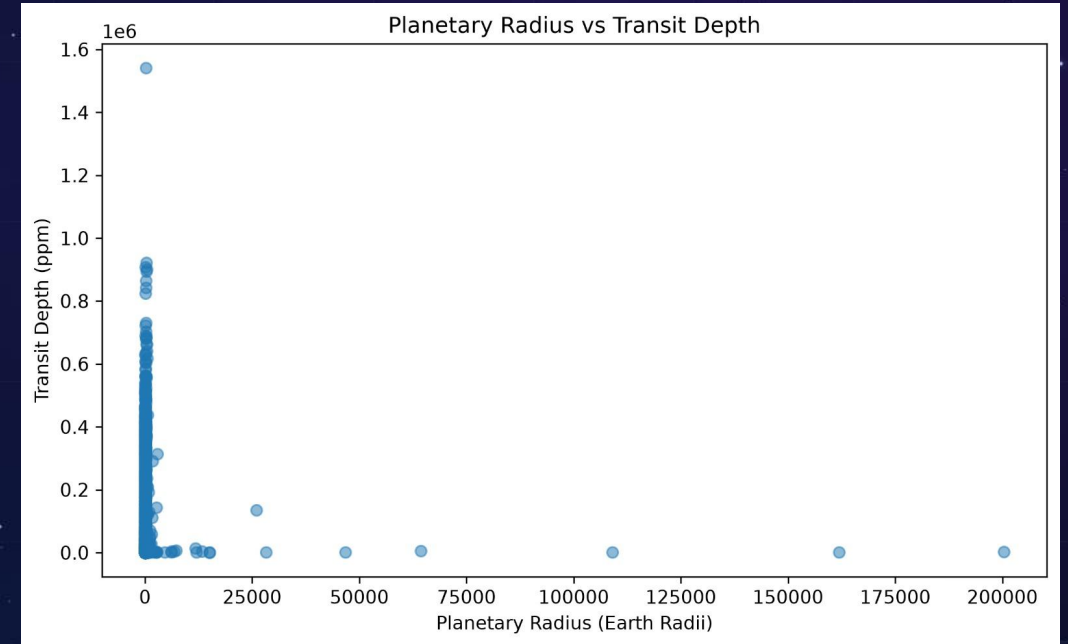
ML classification

We are developing an AI-assisted interactive platform that allows users to ask questions about the NASA Kepler dataset in natural language and receive relevant, understandable information about exoplanet candidates, their host stars, transit events, and ML-based candidate classification and screening.

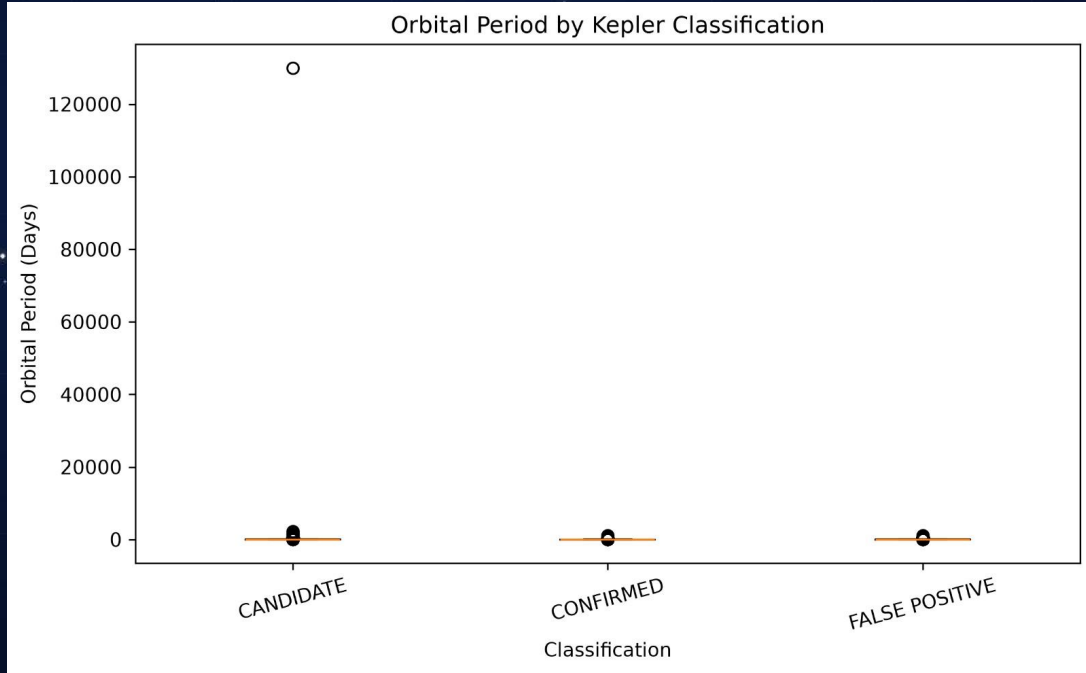
# OUTPUT



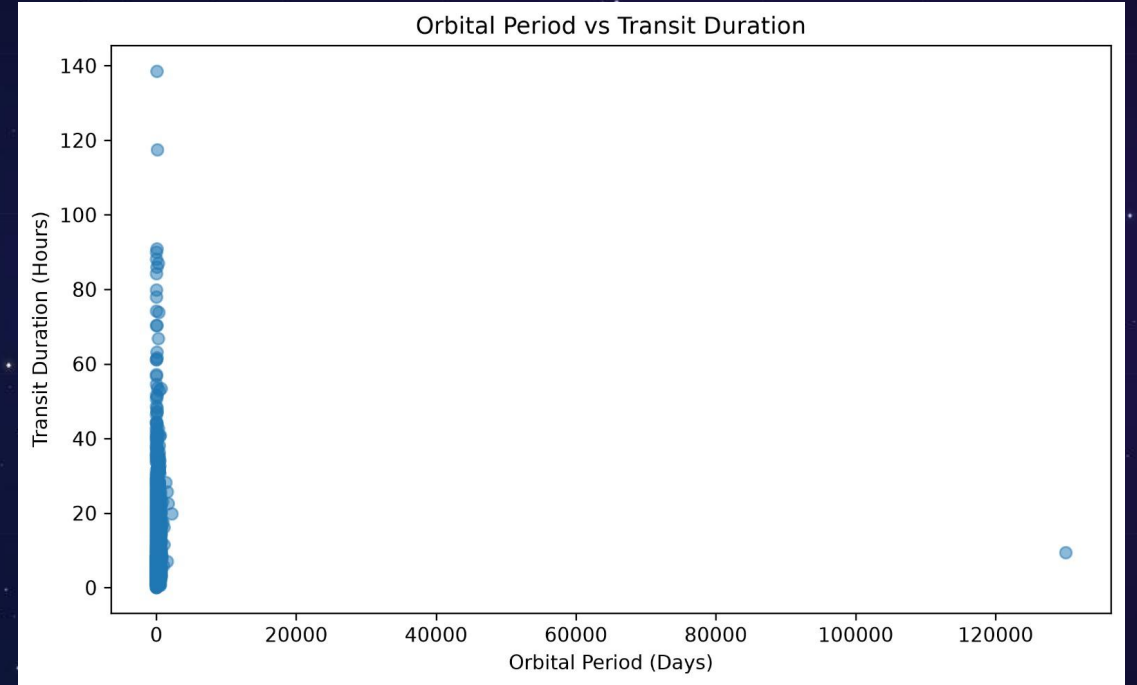
Displays the breakdown of Kepler detected signals, where most objects are flagged as false positives (~4,800), while confirmed exoplanets and unverified candidates each account for around 2,200 to 2,300 objects.



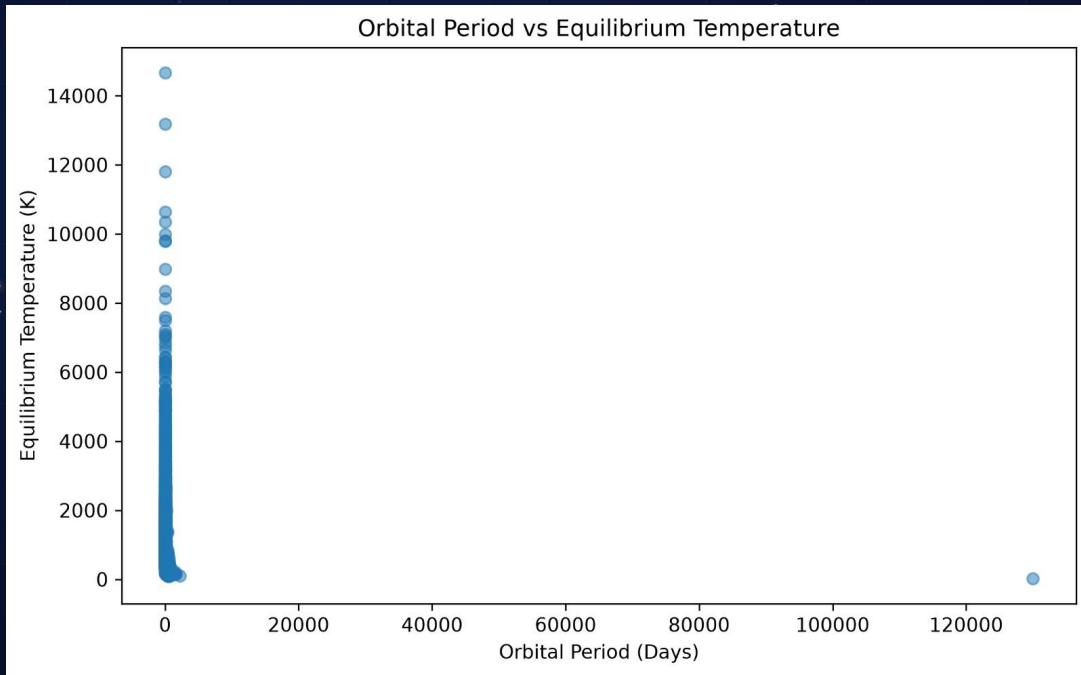
Shows the correlation between an exoplanet's size and how much starlight it blocks, with the vast majority of objects tightly clustered near smaller radii and lower transit depths, alongside a few extreme statistical outliers stretching across larger radii.



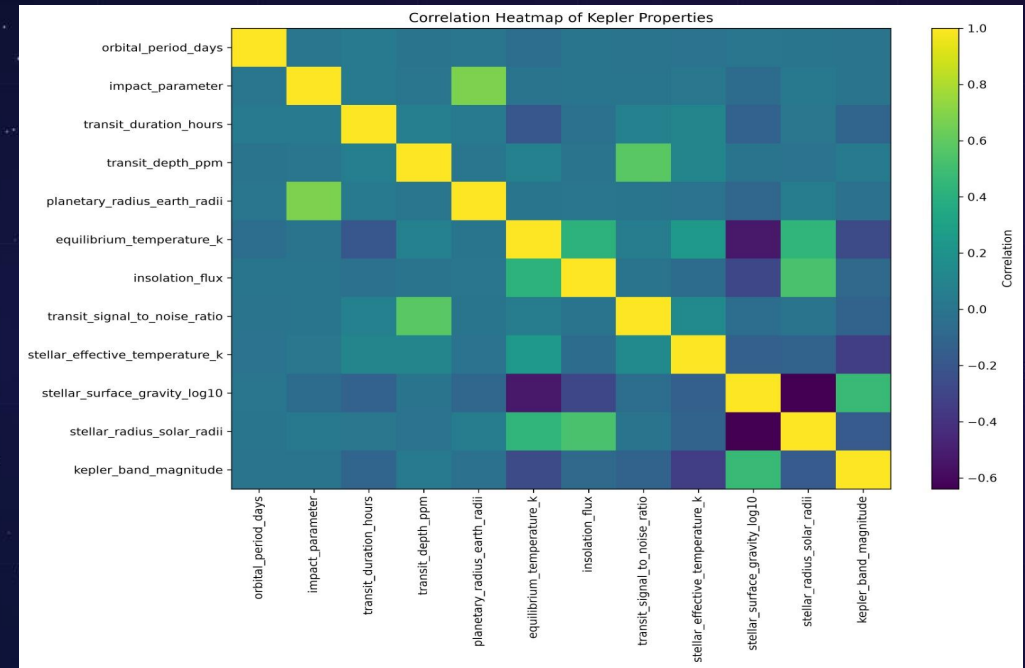
Compares the orbital period distributions across the candidate, confirmed, and false positive categories using a boxplot, showing that almost all detected objects have short orbital periods concentrated near zero days, except for a few extreme outlier candidates exceeding 100,000 days.



Illustrates the relationship between how long an object takes to orbit its star and the time it takes to pass in front of it, revealing that nearly all transits last under 140 hours for short-period orbits, with a single extreme outlier with a very long orbital period (~130,000 days) and short transit duration (~10 hours).



Plots exoplanet temperature against orbital period, showing that objects with very short orbital periods (near zero days) experience extreme temperature ranges (up to  $\sim 14,000$  K), while a far outlier near  $\sim 130,000$  days shows near-zero equilibrium temperature.



Displays pairwise linear correlations across planetary and stellar parameters, highlighting notable positive relationships (such as planetary radius with impact parameter, and transit depth with signal-to-noise ratio) and strong negative relationships (like stellar surface gravity with stellar radius).

# /THANK YOU/

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